

COURTNEY M. CURRIER, Ph.D.

Curriculum Vitae

Postdoctoral Scholar, Stanford University
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RESEARCH INTERESTS

I am an ecosystem ecologist at Stanford University focusing on soil carbon storage potential and persistence in grasslands and savannas. I am interested in understanding how climate change and increasing disturbances, such as fire, are affecting cycling of essential nutrients between plants and soils in dryland ecosystems, and how this underlies changes to carbon dynamics and potential for climate change mitigation across earth's most ubiquitous terrestrial biome. Tools I use are fire and rainfall manipulation experiments in the field, synthesis of spatial datasets, and stable and radioisotopes.

ACADEMIC QUALIFICATIONS

2018–2023 Ph.D. Environmental Life Sciences, Arizona State University, Tempe, AZ
2013–2015 M.Sc. Biology, Arizona State University, Tempe, AZ
2009–2013 B.Sc. Environmental Sciences (Honors Research), University of Notre Dame, Notre Dame, IN

PROFESSIONAL EXPERIENCE

2026–present Postdoctoral Scholar, Doerr School of Sustainability, Stanford University
2023–2026 Postdoctoral Research Associate, Department of Plant Sciences, University of Cambridge, United Kingdom
2025 Lecturer, University of Cambridge
2017–2018 Project Manager, Global Drylands Center, School of Sustainability, Arizona State University
2015–2018 Research Specialist and Lab Manager, Sala Laboratory, Arizona State University

FELLOWSHIPS AND GRANTS

(\$72,290 earned)

2025 Stanford University C-ShaRP Voucher (co-PI)
2023 [rejected] National Science Foundation Postdoctoral Fellowship in Biology: Bridging the gap between ecosystem vulnerability to nitrogen loss and opportunity for uptake
2022 [rejected, semifinalist] Presidential Management Fellowship
2022 PhD Completion Fellowship, School of Life Sciences, Arizona State University
2022 Graduate College Travel Award, Arizona State University
2020 Student Registration Award, Ecological Society of America Biogeosciences Section
2019–2021 Jornada LTER Graduate Research Fellowship

2019	Student Travel Award, Ecological Society of America Southwest Section
2019	Research Training Initiative Grad Student Support Grant, Arizona State University
2018–2019	Arizona State University Graduate College Fellowship
2018	Excellence in Rangeland Ecology Research Travel Award, Ecological Society of America Rangeland Ecology Section
2017	Travel Grant Award, ASU, Graduate and Professional Student Association
2015	Travel Grant Award, ASU, School of Life Sciences
2015	Travel Grant Award, ASU, Graduate and Professional Student Association
2013	[rejected, Honorable Mention] National Science Foundation – Graduate Research Fellowship
2012	National Science Foundation – Research Experience for Undergraduates Fellowship, Niwot Ridge LTER
2012	Travel Grant Award, University of Notre Dame, College of Science
2012	Travel Grant Award, National Science Foundation – Biology REU

HONORS AND AWARDS

2022	[semifinalist] Achievement Rewards for College Scientists
2021	“Grad Slam” Elevator Pitch 3 rd Place Winner, Arizona State University Graduate and Professional Student Association
2020	Outstanding Research Award, Arizona State University Graduate and Professional Student Association
2018–2022	The Graduate College Excellence Award, Arizona State University
2013	Outstanding Environmental Scientist Award, University of Notre Dame
2012–2013	Honors Research Student in Biological Sciences, University of Notre Dame
2011–2012	Dean’s List, University of Notre Dame, College of Science
2011	Clare Boothe Luce Undergraduate Research Award, University of Notre Dame College of Science Summer Research Fellow

PEER-REVIEWED PUBLICATIONS

(H index = 9 | [Google Scholar](#))

Currier, C.M., Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Mkansi, N., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2026. Climate versus fire and herbivory controls on above- and belowground productivity in savannas. In preparation for submission to *Nature Ecology & Evolution*. Full draft available upon request.

Currier, C.M., Reichmann, L.G., and O.E. Sala. 2026. Unresponsive drylands to nitrogen availability: Access to alternative sources? In preparation for submission to *Journal of Geophysical Research: Biogeosciences*. Full draft available upon request.

Powell, R., ..., **Currier C.M.**, ..., et al. 2026. Constraints and opportunities for improved rangeland grazing. In revision, *Nature*.

Currier, C.M., Reichmann, L.G., and O.E. Sala. 2026. Acclimation of the nitrogen cycle to changes in precipitation. *Ecosystems* 29(25). [DOI](#)

Biancari, L., ..., **Currier, C.M.**, ..., et al. 2024.. Drivers of woody dominance across global drylands. *Science Advances* 10,eadn6007. [DOI](#)

- Díaz-Martínez, P., ..., **Currier, C.M.**, ..., et al. 2024. Vulnerability of mineral-associated soil organic carbon to climate across global drylands. *Nature Climate Change* 14: 976–982. [DOI](#)
- Maestre, F., ...**Currier, C.M.**, ..., et. al. 2022. Grazing and ecosystem service delivery in global drylands. *Science* 378: 915-920. [DOI](#)
- Currier, C.M.** and O.E. Sala. 2022. Precipitation versus temperature as phenology controls in drylands. *Ecology*. [DOI](#)
- Osborne, B. B., Bestelmeyer, B. T., **Currier, C. M.**, Homyak, P. M., Throop, H. L., Young, K., and Reed, S. C. 2022. The consequences of climate change for dryland biogeochemistry. *New Phytologist*. [DOI](#)
- Currier, C.M.**, Makings, E., Anderson, J., Slagel, K., and J. Maranville. 2021. Solanaceae Part Seven: Browallia, Calibrachoa, Capsicum, Jaltomata, and Salpichroa. *Canotia* 17: 46-60. [DOI](#)
- Currier, C.M.**, Chaloner, D.T., Ruegg, J., Tiegs, S.D., D’Amore, D., and G.A. Lamberti. 2020. Beyond macronutrient resource subsidies: Pacific salmon (*Oncorhynchus* spp.) as potential vectors of micronutrients. *Aquatic Sciences* 82:50. [DOI](#)
- Sala, O.E., Boone, C.G., Turner II, B.L., and **C.M. Currier**. 2019. The sustainability gap and its implications. *Current Opinion in Environmental Sustainability* 39: 39-43. [DOI](#)
- Delgado-Baquerizo, M., ..., **Currier, C.M.**, ..., et al. 2019. Changes in belowground biodiversity during ecosystem development. *Proceedings of the National Academy of Science* 116(14): 6891-6896. [DOI](#)
- Currier, C.M.**, and J.J. Elser. 2017. Beyond monoculture stoichiometry studies: assessing growth, respiration, and feeding responses of three *Daphnia* species to P-enriched, low C:P lake seston. *Inland Waters* 7(3): 348-357. [DOI](#)
- Rüegg, J., **C.M. Currier**, D.T. Chaloner, S.D. Tiegs, and G.A. Lamberti. 2014. Habitat influences Pacific salmon (*Oncorhynchus* spp.) tissue decomposition in riparian and stream ecosystems. *Aquatic Sciences* 76(4): 623-632. [DOI](#)

SELECTED CONFERENCE PRESENTATIONS

(last five years)

- Currier, C.M.**, Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Mkansi, N., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2025. Fire and aridity controls on soil carbon inputs. Baltimore, MD, USA. August 2025. (talk)
- C.M. Currier** and O.E. Sala. Nitrogen plant uptake in a semiarid ecosystem is modulated by water and plant type. Ecological Society of America Annual Meeting. Montreal, Quebec, Canada. August 2022. (talk)
- C.M. Currier** and O.E. Sala. A temporal perspective of nitrogen cycling in a semiarid grassland under extreme rainfall conditions. Ecological Society of America Annual Meeting. Virtual. August 2021. (talk)
- C.M. Currier** and O.E. Sala. Re-envisioning plant phenology: From data points to painting. Ecological Society of America Annual Meeting. Virtual. August 2020. (Inspire talk, co-organizer of session)

C.M. Currier and O.E. Sala. Extreme precipitation interacts with N cycling in a semi-arid grassland. Ecological Society of America Annual Meeting. Virtual. August 2020. (talk)

INVITED TALKS

(last five years)

Archbold Biological Station, Microbial Ecology Workshop. August 2025.

Emmanuel College, University of Cambridge, College Research Symposium. February 2025.

Department of Plant Sciences, University of Cambridge. February 2025.

University of the Third Age, Botany Seminar. January 2025.

Cambridge Botanical Garden, Science on Sunday Series. July 2024.

OTHER PUBLICATIONS AND ART

Paintings: <https://courtcourrier.github.io/art.html>

C.M. Currier. 2025. “At the cutting edge.” Feature in the *Emmanuel Review*. [Online only](#)

C.M. Currier and T. Feltus. 2023. Finding joy through art, ecology, and bikes. Feature in *The Radavist*. [Online only](#).

C.M. Currier. 2019. Feature in *Phoenix Transect*. [Online only](#)

C.M. Currier and O.E. Sala. 2018. Art provides a new lens for moving through the scientific method. *SciArt Magazine* 33: How can art influence science? [Online only](#)

C.M. Currier. 2018. Feature in *Life Raft Zine*. [Online only](#)

GROUP ART EXHIBITIONS

THIRST 2022: Artists for Humanitarian Aid. [Online only](#). July 2022 (original paintings)

Dimensions of Science. Onyx Gallery, Phoenix, AZ. February 2019 (original painting)

Portraits of Science. MonOrchid Gallery. Phoenix, AZ. May 2017 (original paintings)

SYNERGISTIC ACTIVITIES

2024–present College Research Associate, Emmanuel College, University of Cambridge

2024–present Women in Soil Ecology Mentor

2022 Abstract Reviewer for the Ecological Society of America Southwestern Chapter

2020–2022 Grant Reviewer for the Graduate and Professional Student Association, Arizona State University

2019–2023 Vision Kids Art Program Instructor, Chandler, AZ

2019–2021 Jornada Basin LTER Graduate Student Representative

2019 IsoCamp, University of Utah

2018 Explore National Ecological Observatory Network Workshop, Boulder, CO

2017–present BIODESERT Global Research Group

2017–present Alumni Mentor, Environmental Sciences undergraduate program, Notre Dame

2016–present Asombro Institute, New Mexico State University, Jornada LTER, volunteer

2013–2014 Arizona State University, School of Life Sciences Graduate Executive Committee, secretary

2013–present Arizona State University Ask a Biologist, volunteer

2012–2013 Notre Dame Biological Sciences Senior Leadership Committee

TEACHING

University of Cambridge Lecturer and Supervisor
Part II Plant Sciences Module: Global Change

Graduate Teaching Assistant at Arizona State University
(*indicates significant role in course development)

Biology 426: Limnology*

Biology 320: Fundamentals of Ecology*

Biology 281: General Biology I for Majors

Biology 282: General Biology II for Majors*

Biology 181: General Biology I

Biology 182: General Biology II

Biology 100: The Living World

Undergraduate Teaching Assistant at Notre Dame
Biology 30312: General Ecology